

Treatment of Neurovisceral Attacks in the Acute Porphyrrias

Most patients with acute porphyria experiencing an attack can now be treated successfully, and life-threatening complications such as muscle paralysis, respiratory failure, and coma can be avoided if appropriate aggressive treatment is implemented immediately. Untreated or inappropriately treated patients experiencing an acute porphyric attack may have a 10 percent to 15 percent risk of mortality. Today, with mechanism-based therapeutic intervention, the mortality rate of the acute attack has decreased to approximately 2 percent.

There are four major steps in treating an acute porphyria attack.

1. Identification and Elimination of Precipitating Factors

All potentially porphyrinogenic drugs should be stopped immediately. The patient should be monitored carefully and admitted to an intensive care unit if possible. Alcohol, hormonal fluctuations (e.g., from oral contraceptives or pregnancy), fasting, and infections are common triggers and should be addressed.

Drug safety information is available from reliable sources such as the European Porphyria Initiative (EPI) at www.porphyrria-europe.com, which provides up-to-date lists of safe and unsafe medications for AIP patients.

2. Symptomatic Management of Neurovisceral Manifestations

Prompt symptomatic treatment is essential for improving patient comfort and preventing complications.

Symptom/Sign	Recommended Treatment
Abdominal pain	Acetylsalicylic acid; morphine
Vomiting	Chlorpromazine; promethazine
Hypertension and tachycardia	Propranolol
Hyponatremia	Electrolyte replacement; saline infusions
Convulsions	Gabapentin
Constipation	Neostigmine

Note: Opioids like morphine are safe and effective for pain control. Pethidine (meperidine) may also be used but with caution due to seizure risk at high doses.

3. Carbohydrate Loading

Intravenous glucose loading is a cornerstone of initial therapy. Administer **400–500 g of glucose per 24 hours** using 5% or 10% glucose solutions. This suppresses hepatic heme synthesis and reduces accumulation of toxic porphyrin precursors.

Concurrent administration of **furosemide** helps prevent fluid overload while maintaining diuresis. Serum electrolytes—especially sodium—must be closely monitored due to the risk of hyponatremia.

4. Specific Heme Therapy

If symptoms persist beyond 24–48 hours despite carbohydrate loading and supportive care, specific treatment with intravenous heme is indicated:

- **Heme arginate (Normosang®)** – 3 mg/kg body weight daily for 4 days (used in Europe)
- **Hematin (Panhematin®)** – 3–4 mg/kg body weight daily for 4 days (used in the USA)

These agents replete hepatic heme stores, downregulate δ -aminolevulinic acid synthase (ALAS1), and reduce production of neurotoxic precursors (ALA and PBG).

Monitoring and Laboratory Assessment

- Measure urinary **δ-aminolevulinic acid (ALA)** and **porphobilinogen (PBG)** excretion daily during the acute phase.
- Monitor renal function, electrolytes, neurological status, and fluid balance closely.

Additional Considerations

- **Contact a specialized porphyria center** early for guidance on diagnosis, treatment, and long-term management.
- Patients should wear **medical warning bracelets** and carry a list of drugs to avoid.
- Drugs considered safe for use in acute porphyria are listed in standard reference tables (e.g., Table 132-10).

Box 132-2: Treatment of the Acute Porphyrias

TYPE

Acute intermittent porphyria, variegate porphyria, hereditary coproporphyria, δ-aminolevulinic acid dehydratase deficiency porphyria

MANAGEMENT

Acute Attacks - Identify and eliminate precipitating factors (e.g., porphyrinogenic drugs, alcohol, hormones) - Monitor in ICU and/or contact a specialized porphyria center - Provide appropriate analgesia (e.g., pethidine or other opiates) - Manage nausea and vomiting (e.g., promazine, chlorpromazine, triflupromazine) - Administer intravenous heme arginate (3 mg/kg/day × 4 days) or hematin (3–4 mg/kg/day × 4 days) - If needed, initiate intravenous glucose loading - Measure urinary ALA and PBG excretion daily

Variagate Porphyria and Hereditary Coproporphyria – Cutaneous

Manifestations - Photoprotection using broadband sunscreens and protective

clothing - Avoid sunlight exposure and skin trauma